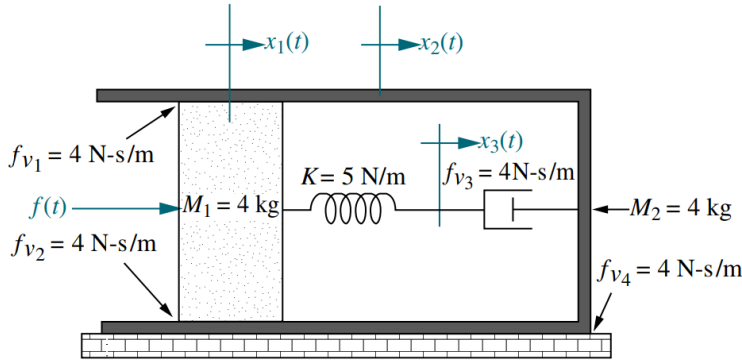


Chapter 2B Homework

22, 25, 28, 29, 32, 33, 35, 37, 50, 51, 52

Question 22(a)



At the $M_1 = 4 \text{ kg}$ mass, the equation of motion is:

$$(M_1 s^2 + f_{v1} s + f_{v2} s + K)X_1(s) - (f_{v1} s + f_{v2} s)X_2(s) - KX_3(s) = F(s)$$

$$(4s^2 + 8s + 5)X_1(s) - 8sX_2(s) - 5X_3(s) = F(s) \dots [1]$$

At the $M_2 = 4 \text{ kg}$ mass, the equation of motion is:

$$-(f_{v1} s + f_{v2} s)X_1(s) + (M_2 s^2 + f_{v1} s + f_{v2} s + f_{v3} s + f_{v4} s)X_2(s) - f_{v3} sX_3(s) = 0$$

$$-8sX_1(s) + (4s^2 + 16s)X_2(s) - 4sX_3(s) = 0 \dots [2]$$

Assume that at $x_3(t)$, there is a mass $M_3 = 0 \text{ kg}$. The equation of motion is:

$$-KX_1(s) - f_{v3}X_2(s) + (M_3 s^2 + f_{v3} s + K)X_3(s) = 0$$

$$-5X_1(s) - 4sX_2(s) + (4s + 5)X_3(s) = 0 \dots [3]$$

The system of equations from [1] – [3] are:

$$\begin{bmatrix} (4s^2 + 8s + 5) & -8s & -5 \\ -8s & (4s^2 + 16s) & -4s \\ -5 & -4s & (4s + 5) \end{bmatrix} \begin{bmatrix} X_1(s) \\ X_2(s) \\ X_3(s) \end{bmatrix} = \begin{bmatrix} F(s) \\ 0 \\ 0 \end{bmatrix}$$

The determinant of the system Δ is:

$$\Delta = \begin{vmatrix} (4s^2 + 8s + 5) & -8s & -5 \\ -8s & (4s^2 + 16s) & -4s \\ -5 & -4s & (4s + 5) \end{vmatrix}$$

$$\Delta = 64s^5 + 400s^4 + 688s^3 + 240s^2$$

Substitute the right-hand side into the third column for $X_3(s)$, the determinant Δ_3 is:

$$\Delta_3 = \begin{vmatrix} (4s^2 + 8s + 5) & -8s & F(s) \\ -8s & (4s^2 + 16s) & 0 \\ -5 & -4s & 0 \end{vmatrix}$$

$$\Delta_3 = (52s^2 + 80s)F(s)$$

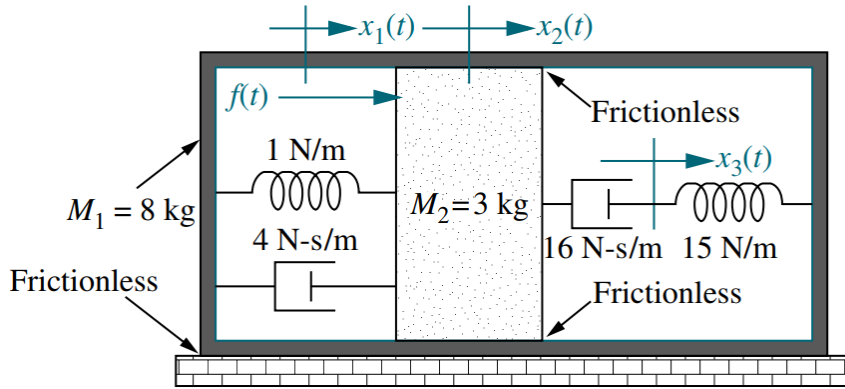
Solving for $X_3(s)/F(s)$ using:

$$X_3(s) = \frac{\Delta_3}{\Delta}$$

$$X_3(s) = \frac{(52s^2 + 80s)F(s)}{64s^5 + 400s^4 + 688s^3 + 240s^2}$$

$$\frac{X_3(s)}{F(s)} = \frac{13s + 20}{4s(4s^3 + 25s^2 + 43s + 15)}$$

Question 22(b)



At the $M_1 = 8$ kg mass, the equation of motion is:

$$(8s^2 + 4s + 1 + 15)X_1(s) - (4s + 1)X_2(s) - 15X_3(s) = 0 \dots [1]$$

At the $M_2 = 3$ kg mass, the equation of motion is:

$$-(4s + 1)X_1(s) + (3s^2 + 4s + 16s + 1)X_2(s) - 16sX_3(s) = F(s) \dots [2]$$

Assume that at $x_3(t)$, there is a mass $M_3 = 0$ kg. The equation of motion is:

$$-15X_1(s) - 16sX_2(s) + (16s + 15)X_3(s) = 0 \dots [3]$$

The system of equations from [1] – [3] are:

$$\begin{bmatrix} (8s^2 + 4s + 16) & -(4s + 1) & -15 \\ -(4s + 1) & (3s^2 + 20s + 1) & -16s \\ -15 & -16s & (16s + 15) \end{bmatrix} \begin{bmatrix} X_1(s) \\ X_2(s) \\ X_3(s) \end{bmatrix} = \begin{bmatrix} 0 \\ F(s) \\ 0 \end{bmatrix}$$

The determinant of the system Δ is:

$$\Delta = \begin{vmatrix} (8s^2 + 4s + 16) & -(4s + 1) & -15 \\ -(4s + 1) & (3s^2 + 20s + 1) & -16s \\ -15 & -16s & (16s + 15) \end{vmatrix}$$

$$\Delta = 384s^5 + 1064s^4 + 3476s^3 + 165s^2$$

Substitute the right-hand side into the third column for $X_3(s)$, the determinant Δ_3 is:

$$\Delta_3 = \begin{vmatrix} (8s^2 + 4s + 16) & -(4s + 1) & 0 \\ -(4s + 1) & (3s^2 + 20s + 1) & F(s) \\ -15 & -16s & 0 \end{vmatrix}$$

$$\Delta_3 = (128s^3 + 64s^2 + 316s + 15)F(s)$$

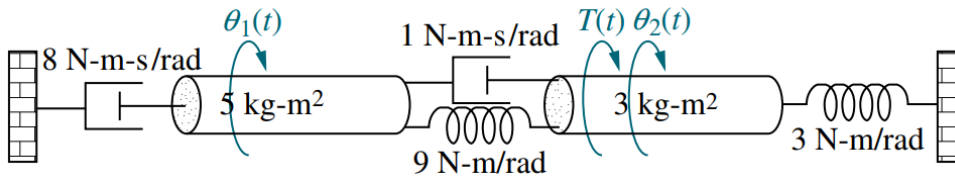
Solving for $X_3(s)/F(s)$ using:

$$X_3(s) = \frac{\Delta_3}{\Delta}$$

$$X_3(s) = \frac{(128s^3 + 64s^2 + 316s + 15)F(s)}{384s^5 + 1064s^4 + 3476s^3 + 165s^2}$$

$$\frac{X_3(s)}{F(s)} = \frac{128s^3 + 64s^2 + 316s + 15}{s^2(384s^3 + 1064s^2 + 3476s + 165)}$$

Question 25(a)



At the 5 kg-m² inertia, the equation of motion is:

$$(5s^2 + 8s + s + 9)\theta_1(s) - (s + 9)\theta_2(s) = 0$$

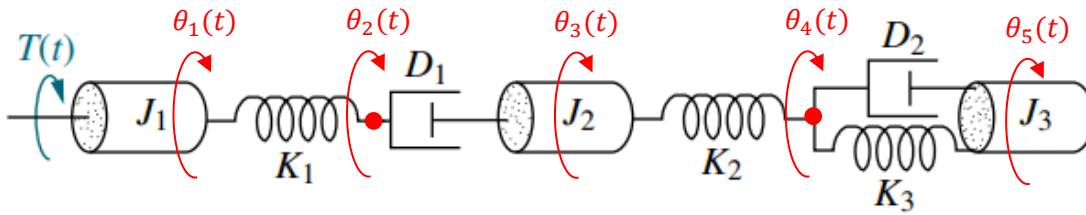
$$(5s^2 + 9s + 9)\theta_1(s) - (s + 9)\theta_2(s) = 0$$

At the 3 kg-m² inertia, the equation of motion is:

$$-(s + 9)\theta_1(s) + (3s^2 + s + 9 + 3)\theta_2(s) = T(s)$$

$$-(s + 9)\theta_1(s) + (3s^2 + s + 12)\theta_2(s) = T(s)$$

Question 25(b)



There are five degrees of freedom, therefore, there are four equations of motion.

At θ_1 :

$$(J_1s^2 + K_1)\theta_1(s) - K_1\theta_2(s) = T(s)$$

At θ_2 :

$$-K_1\theta_1(s) + (D_1s + K_1)\theta_2(s) - D_1s\theta_3(s) = 0$$

At θ_3 :

$$-D_1s\theta_2(s) + (J_2s^2 + D_1s + K_2)\theta_3(s) - K_2\theta_4(s) = 0$$

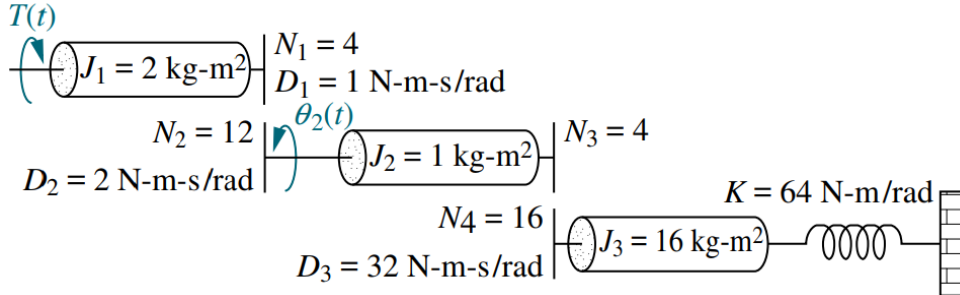
At θ_4 :

$$-K_2\theta_3(s) + (D_2s + (K_2 + K_3))\theta_4(s) - (D_2s + K_3)\theta_5(s) = 0$$

At θ_5 :

$$-(D_2s + (K_2 + K_3))\theta_4(s) + (J_3s^2 + D_2s + K_3)\theta_5(s) = 0$$

Question 28



Reflect all the impedances to $\theta_2(s)$ to solve for $\theta_2(s)/T(s)$. The equivalent inertia at $\theta_2(s)$ is:

$$J_e = J_1 \left(\frac{N_2}{N_1} \right)^2 + J_2 + J_3 \left(\frac{N_3}{N_4} \right)^2 = 2 \left(\frac{12}{4} \right)^2 + 1 + 16 \left(\frac{4}{16} \right)^2$$

$$J_e = 20 \text{ kg-m}^2$$

The equivalent rotational damping at $\theta_2(s)$ is:

$$D_e = D_1 \left(\frac{N_2}{N_1} \right)^2 + D_2 + D_3 \left(\frac{N_3}{N_4} \right)^2 = 1 \left(\frac{12}{4} \right)^2 + 2 + 32 \left(\frac{4}{16} \right)^2$$

$$D_e = 13 \text{ N-m-s/rad}$$

The equivalent rotational spring at $\theta_2(s)$ is:

$$K_e = K \left(\frac{N_3}{N_4} \right)^2 = 64 \left(\frac{4}{16} \right)^2$$

$$K_e = 4 \text{ N-m/rad}$$

The equivalent torque applied at $\theta_2(s)$ is:

$$T_e(s) = T(s) \left(\frac{N_2}{N_1} \right) = T(s) \left(\frac{12}{4} \right)$$

$$T_e(s) = 3T(s)$$

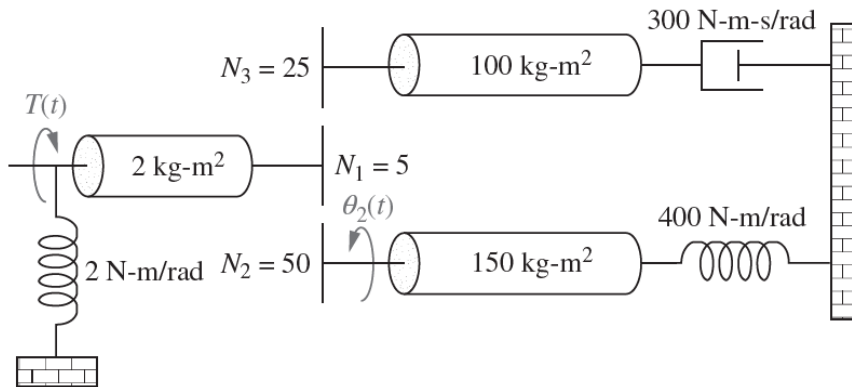
The equation of motion at $\theta_2(s)$ is:

$$(J_e s^2 + D_e s + K_e) \theta_2(s) = T_e(s)$$

$$(20s^2 + 13s + 4) \theta_2(s) = 3T(s)$$

$$\frac{\theta_2(s)}{T(s)} = \frac{3}{20s^2 + 13s + 4}$$

Question 29



Reflecting impedances across gears from the right hand side to the left hand side one gets:

$$J_{eq} = 2 + 100 \left(\frac{5}{25} \right)^2 + 150 \left(\frac{5}{50} \right)^2 = 7.5$$

$$D_{eq} = 300 \left(\frac{5}{25} \right)^2 = 12$$

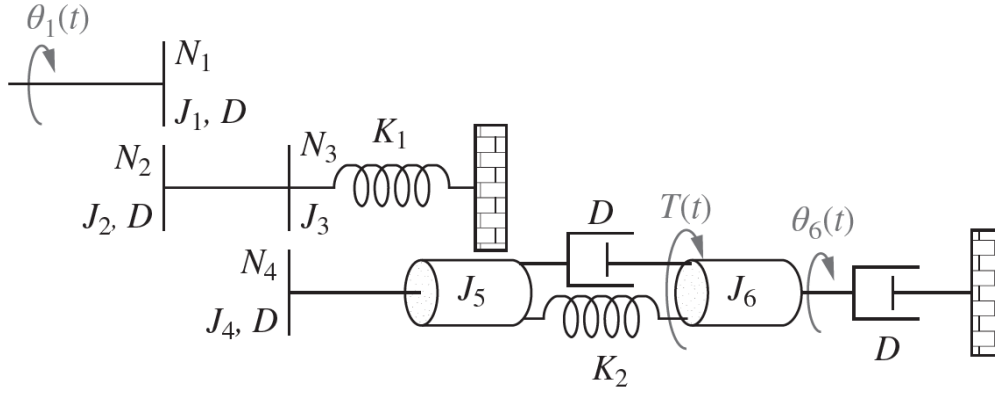
$$K_{eq} = 2 + 400 \left(\frac{5}{50} \right)^2 = 6$$

So $(7.5s^2 + 12s + 6)\theta(s) = T(s)$.

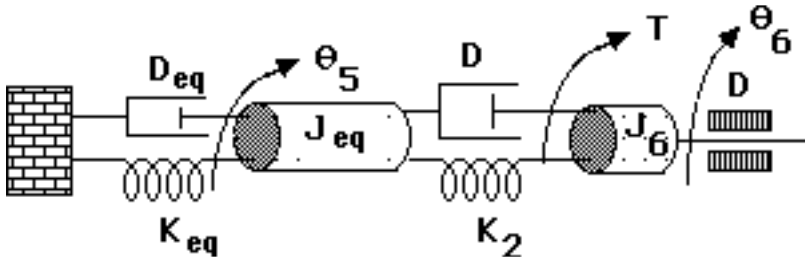
Since $\frac{\theta}{\theta_2} = \frac{N_2}{N_1} = 10$, $(7.5s^2 + 12s + 6) 10 \theta_2(s) = T(s)$

$$\frac{\theta_2(s)}{T(s)} = \frac{1}{75s^2 + 120s + 60} = \frac{0.0133}{s^2 + 1.6s + 0.8}$$

Question 32



Reflect impedances to the left of J5 to J5 and obtain the following equivalent circuit:



Writing the equations of motion,

$$[J_{eq}s^2 + (D_{eq} + D)s + (K_2 + K_{eq})]\theta_5(s) - [Ds + K_2]\theta_6(s) = 0$$

$$-[Ds + K_2]\theta_5(s) + [J_6s^2 + 2Ds + K_2]\theta_6(s) = T(s)$$

From the first equation,

$$\frac{\theta_6(s)}{\theta_5(s)} = \frac{J_{eq}s^2 + (D_{eq} + D)s + (K_2 + K_{eq})}{Ds + K_2}$$

But

$$\frac{\theta_5(s)}{\theta_1(s)} = \frac{N_1 N_3}{N_2 N_4}$$

Therefore,

$$\frac{\theta_6(s)}{\theta_1(s)} = \frac{N_1 N_3}{N_2 N_4} \left(\frac{J_{eq}s^2 + (D_{eq} + D)s + (K_2 + K_{eq})}{Ds + K_2} \right)$$

where,

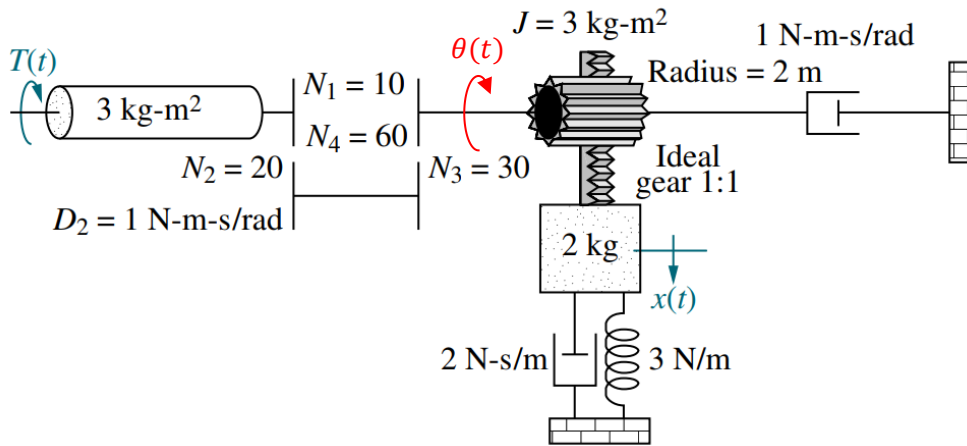
$$J_{eq} = J_1 \left(\frac{N_4 N_2}{N_3 N_1} \right)^2 + (J_2 + J_3) \left(\frac{N_4}{N_3} \right)^2 + (J_4 + J_5)$$

$$K_{eq} = K_1 \left(\frac{N_4}{N_3} \right)^2$$

and

$$D_{eq} = D \left[\left(\frac{N_4 N_2}{N_3 N_1} \right)^2 + \left(\frac{N_4}{N_3} \right)^2 + 1 \right]$$

Question 33



Reflect all the impedances and the applied torque to the centre at $\theta(s)$, the equivalent impedances at $\theta(s)$ are:

$$J_e = 3 + 3 \left(\frac{N_2}{N_1} \right)^2 \left(\frac{N_4}{N_3} \right)^2 = 3 + 3 \left(\frac{20}{10} \times \frac{60}{30} \right)^2 = 51 \text{ kg-m}^2$$

$$D_e = 1 + 1 \left(\frac{N_4}{N_3} \right)^2 = 1 + 1 \left(\frac{60}{30} \right)^2 = 5 \text{ N-m-s/rad}$$

$$K_e = 0 \text{ N-m/rad}$$

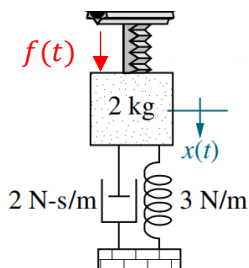
$$T_e(s) = T(s) \left(\frac{N_2}{N_1} \right) \left(\frac{N_4}{N_3} \right) = T(s) \left(\frac{20}{10} \times \frac{60}{30} \right) = 4T(s)$$

The equation of motion at the gear $\theta(s)$ is:

$$(J_e s^2 + D_e s + K_e) \theta(s) + rF(s) = T_e(s)$$

$$(51s^2 + 5s) \theta(s) + 2F(s) = 4T(s) \dots [1]$$

where $rF(s)$ in [1] is the torque required to move the 2 kg mass. Isolate the translation part of the system:



$f(t)$ is the translational force applied by the gear. The equation of motion of the mass is:

$$(2s^2 + 2s + 3)X(s) = F(s) \dots [2]$$

Substitute [2] into [1]:

$$(51s^2 + 5s) \theta(s) + 2(2s^2 + 2s + 3)X(s) = 4T(s) \dots [3]$$

The relation between the translation motion and the rotational motion is:

$$X(s) = r\theta(s) = 2\theta(s)$$

$$\theta(s) = \frac{X(s)}{2} \dots [4]$$

Substitute [4] into [3] and solve for $X(s)/T(s)$:

$$(51s^2 + 5s) \frac{X(s)}{2} + 2(2s^2 + 2s + 3)X(s) = 4T(s)$$

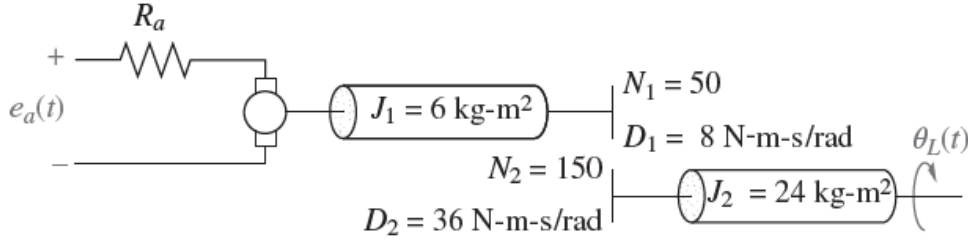
$$\left(\frac{51}{2}s^2 + \frac{5}{2}s\right)X(s) + (4s^2 + 4s + 6)\theta(s) = 4T(s)$$

$$\left(\frac{59}{2}s^2 + \frac{13}{2}s + 6\right)X(s) = 4T(s) \dots \times 2$$

$$(59s^2 + 13s + 12)X(s) = 8T(s)$$

$$\frac{X(s)}{T(s)} = \frac{8}{59s^2 + 13s + 12}$$

Question 34



Reflecting through gears the inertia and damping from the load side to motor shaft one gets,

$$J_m = 6 + 24 \left(\frac{50}{150}\right)^2 = 8.667 \text{ and } D_m = 8 + 36 \left(\frac{50}{150}\right)^2 = 54$$

Note from the motor load curve that $\frac{K_t}{R_a} = \frac{T_{stall}}{e_a} = \frac{150}{60} = 2.5$ and $K_b = \frac{e_a}{\omega_{no-load}} = \frac{60}{100} = \frac{3}{5}$.

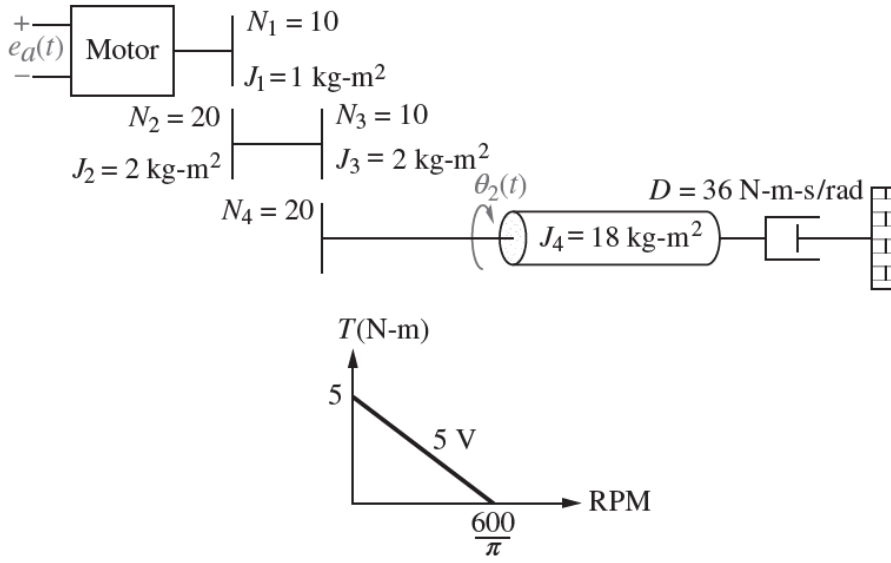
Substituting all of the above, one gets (see equation development in Chapter 2.8)

$$\frac{\theta_m}{E_a} = \frac{\frac{K_t}{R_a J_m}}{s \left(s + \frac{1}{J_m} \left(D_m + \frac{K_t K_b}{R_a} \right) \right)} = \frac{0.2885}{s(s + 6.4036)}$$

Noting that $\frac{\theta_m}{\theta_L} = \frac{N_2}{N_1} = 3$

$$\frac{\theta_L}{E_a} = \frac{0.09615}{s(s + 6.4036)}$$

Question 35



The parameters are:

$$\frac{K_t}{R_a} = \frac{T_s}{E_a} = \frac{5}{5} = 1$$

$$K_b = \frac{E_a}{\omega} = \frac{5}{\frac{600}{\pi} \cdot 2\pi \frac{1}{60}} = \frac{1}{4}$$

$$J_m = 18 \left(\frac{1}{4}\right)^2 + 4 \left(\frac{1}{2}\right)^2 + 1 = 3.125$$

$$D_m = 36 \left(\frac{1}{4}\right)^2 = 2.25$$

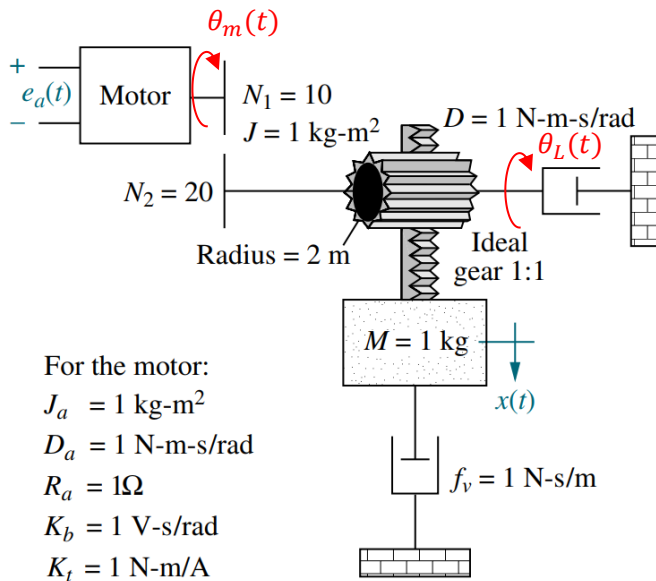
Thus,

$$\frac{\theta_m(s)}{E_a(s)} = \frac{\frac{1}{3.125}}{s \left[s + \frac{1}{3.125} \left(2.25 + (1) \left(\frac{1}{4} \right) \right) \right]} = \frac{0.32}{s(s + 0.8)}$$

Since: $\theta_2(s) = \frac{1}{4} \theta_m(s)$; then:

$$\frac{\theta_2(s)}{E_a(s)} = \frac{0.08}{s(s + 0.8)}$$

Question 37



The equivalent impedances reflected to $\theta_L(s)$ are:

$$J_e = J + J_a \left(\frac{N_2}{N_1} \right)^2 = 1 + 1 \left(\frac{20}{10} \right)^2 = 5 \text{ kg-m}^2$$

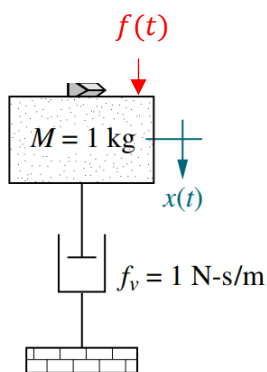
$$D_e = D + D_a \left(\frac{N_2}{N_1} \right)^2 = 1 + 1 \left(\frac{20}{10} \right)^2 = 5 \text{ N-m-s/rad}$$

The equation of motion at the gear $\theta_L(s)$ is:

$$(J_e s^2 + D_e s) \theta_L(s) + rF(s) = T_{eq}(s)$$

$$(5s^2 + 5s) \theta_L(s) + 2F(s) = T_{eq}(s) \dots [1]$$

$T_{eq}(s)$ is the equivalent torque produced by the gear and $rF(s)$ is the torque required to move the mass. Isolate the translation part of the system:



$f(t)$ is the translational force applied by the gear. The equation of motion of the mass is:

$$(s^2 + s)X(s) = F(s) \dots [2]$$

Substitute [2] into [1]:

$$(5s^2 + 5s) \theta_L(s) + 2(s^2 + s)X(s) = T_{eq}(s) \dots [3]$$

The relation between the translation motion and the rotational motion is:

$$X(s) = r\theta_L(s) = 2\theta_L(s) \dots [4]$$

Substitute [4] into [3]:

$$(5s^2 + 5s)\theta_L(s) + 2(s^2 + s)(2\theta_L(s)) = T_{eq}(s)$$

$$(9s^2 + 9s)\theta_L(s) = T_{eq}(s) \dots [5]$$

[5] is the equation of motion at the gear and contains the equivalent impedances at the gear $\theta_L(s)$. This is done so that the translation part of the system is included into the equation of motion. Reflect these impedances to the motor $\theta_m(s)$ to obtain the equivalent impedances seen by the armature of the motor:

$$J_a = J_L \left(\frac{N_1}{N_2} \right)^2 = 9 \left(\frac{10}{20} \right)^2 = 2.25 \text{ kg-m}^2$$

$$D_a = D_L \left(\frac{N_2}{N_1} \right)^2 = 9 \left(\frac{10}{20} \right)^2 = 2.25 \text{ N-m-s/rad}$$

The transfer function for a DC motor is:

$$\frac{\theta_m(s)}{E_a(s)} = \frac{\frac{K_t}{R_a J_a}}{s \left(s + \frac{1}{J_a} \left(D_a + \frac{K_t}{R_a} K_b \right) \right)}$$

$$\frac{\theta_m(s)}{E_a(s)} = \frac{\frac{1}{2.25}}{s \left(s + \frac{1}{2.25} (2.25 + 1) \right)}$$

$$\frac{\theta_m(s)}{E_a(s)} = \frac{0.44}{s(s + 1.44)} \dots [6]$$

To obtain $X(s)/E_a(s)$, we need to relate $\theta_L(s)$ to $\theta_m(s)$ and use [4] to relate $\theta_L(s)$ in terms of $X(s)$:

$$\theta_L(s) = \left(\frac{N_1}{N_2} \right) \theta_m(s) = \frac{1}{2} \theta_m(s)$$

$$\theta_m(s) = 2\theta_L(s) \dots [7]$$

Substitute [7] into [6]:

$$\frac{\theta_L(s)}{E_a(s)} = \frac{0.44}{2s(s + 1.44)} \dots [8]$$

Substitute [4] into [8]:

$$\frac{X(s)}{E_a(s)} = \frac{2(0.44)}{2s(s + 1.44)} = \frac{0.44}{s(s + 1.44)}$$

Question 50(a)

Expressing $\frac{A_e}{A} \sqrt{2gh}$ as a Taylor series around h_0

$$\frac{A_e}{A} \sqrt{2gh} \approx \frac{A_e}{A} \sqrt{2gh_0} + \frac{\partial}{\partial h} \left(\frac{A_e}{A} \sqrt{2gh} \right)_{h_0} \partial h = \frac{A_e}{A} \sqrt{2gh_0} + \frac{A_e}{A} \frac{g}{\sqrt{2gh_0}} \partial h \dots [1]$$

Also, taking ∂h and ∂q as deviation variables

$$h = h_0 + \partial h \dots [2]$$

and

$$q = q_0 + \partial q \dots [3]$$

Substituting [1], [2], and [3] into the given nonlinear equation and eliminating the equilibrium values yields the linear equation

$$\frac{d\partial h}{dt} + \frac{A_e}{A} \frac{g}{\sqrt{2gh_0}} \partial h = \frac{\partial q}{\rho A}$$

Thus, the transfer function is

$$\frac{H(s)}{Q(s)} = \frac{1/\rho A}{\left(s + \frac{A_e}{A} \frac{g}{\sqrt{2gh_0}}\right)}$$

Question 50(b)

Substituting $q_e = \rho A_e \sqrt{2gh_{av}}$ into $e_1 q - e q_e = \frac{d}{dt}(\rho A e h_{av})$

$$e_1 q - e \rho A_e \sqrt{2gh_{av}} = \rho A h_{av} \frac{de}{dt}$$

Rearranging,

$$\rho A h_{av} \frac{de}{dt} + e \rho A_e \sqrt{2gh_{av}} = e_1 q$$

Simplifying,

$$\frac{de}{dt} + \frac{A_e}{A h_{av}} \sqrt{2gh_{av}} e = e_1 q$$

Taking the Laplace transform

$$sE(s) + \frac{A_e}{A h_{av}} \sqrt{2gh_{av}} E(s) = e_1 Q(s)$$

From which,

$$\frac{E(s)}{Q(s)} = \frac{e_1}{\left(s + \frac{A_e}{A h_{av}} \sqrt{2gh_{av}}\right)}$$

Question 51(a)

The first two equations are nonlinear because of the Tv products on their right hand side. Otherwise the equations are linear.

Question 51(b)

To find the equilibria let

$$\frac{dT}{dt} = \frac{dT^*}{dt} = \frac{dv}{dt} = 0$$

Leading to

$$s - dT - \beta T v = 0$$

$$\beta T v - \mu T^* = 0$$

$$kT^* - cv = 0$$

The first equilibrium is found by direct substitution. For the second equilibrium, solve the last two equations for T^*

$$T^* = \frac{\beta T v}{\mu}$$

and

$$T^* = \frac{c v}{k}$$

Equating we get that $T = \frac{c\mu}{\beta k}$

Substituting the latter into the first equation after some algebraic manipulations we get that

$$v = \frac{k s}{c \mu} - \frac{d}{\beta}$$

It follows that

$$T^* = \frac{c v}{k} = \frac{s}{\mu} - \frac{c d}{k \beta}$$

Question 52(a)

From

$$a = \frac{F - F_w}{k_m \cdot m}$$

we have:

$$F = F_w + k_m \cdot m \cdot a = F_{RO} + F_L + F_{St} + k_m \cdot m \cdot a \dots [1]$$

Substituting for the motive force, F , and the resistances F_{RO} , F_L , and F_{st} using the equations given in the problem, yields the equation:

$$F = \frac{P \cdot \eta_{tot}}{v} = f \cdot m \cdot g \cdot \cos \alpha + m \cdot g \cdot \sin \alpha + 0.5 \cdot \rho \cdot C_w \cdot A \cdot (v + v_{hw})^2 + k_m \cdot m \cdot a \dots [2]$$

Question 52(b)

Noting that constant acceleration is assumed, the average values for speed and acceleration are:

$$a_{av} = 20 \text{ (km/h)} / 4 \text{ s} = 5 \text{ km/h.s} = 5 \times 1000 / 3600 \text{ m/s}^2 = 1.389 \text{ m/s}^2$$

$$v_{av} = 50 \text{ km/h} = 50,000 / 3,600 \text{ m/s} = 13.89 \text{ m/s}$$

The motive force, F (in N), and power, P (in kW) can be found from eq. [2]:

$$F_{av} = 0.011 \times 1590 \times 9.8 + 0.5 \times 1.2 \times 0.3 \times 2 \times 13.89^2 + 1.2 \times 1590 \times 1.389 = 2891 \text{ N}$$

$$P_{av} = F_{av} \cdot v / \eta_{tot} = 2891 \times 13.89 / 0.9 = 44,617 \text{ N.m/s} = 44.62 \text{ kW}$$

To maintain a speed of 60 km/h while climbing a hill with a gradient $\alpha = 5^\circ$, the car engine or motor needs to overcome the climbing resistance:

$$F_{St} = m \cdot g \cdot \sin \alpha = 1590 \cdot 9.8 \cdot \sin 5^\circ = 1358 \text{ N}$$

Thus, the additional power, P_{add} , the car needs after reaching 60 km/h to maintain its speed while climbing a hill with a gradient $\alpha = 5^\circ$ is:

$$P_{add} = F_{St} \cdot v / \eta = 1358 \times 60 \times 1000 / (3,600 \times 0.9) = 25,149 \text{ W} = 25.15 \text{ kW}$$

Question 52(c)

Substituting for the car parameters into equation [2] yields:

$$F = 0.011 \times 1590 \times 9.8 + 0.5 \times 1.2 \times 0.3 \times 2 v^2 + 1.2 \times 1590 dv/dt$$

or

$$F(t) = 171.4 + 0.36 v^2 + 1908 dv/dt \dots [3]$$

To linearize this equation about $v_o = 50 \text{ km/h} = 13.89 \text{ m/s}$, we use the truncated Taylor series:

$$v^2 - v_o^2 \approx \frac{d(v^2)}{dv} \bigg|_{v=v_o} (v - v_o) = 2v_o \cdot (v - v_o) \dots [4]$$

from which we obtain:

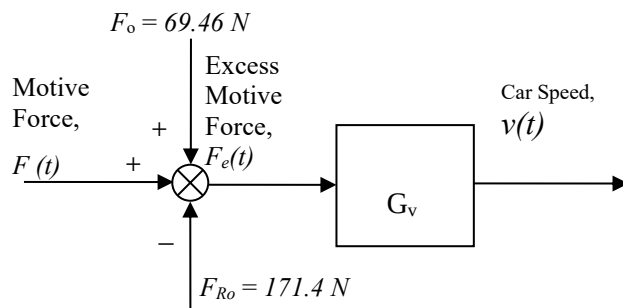
$$v^2 = 2v_o \cdot v - v_o^2 = 27.78 \cdot v - 13.89^2 \dots [5]$$

Substituting from equation [5] into [3] yields:

$$F(t) = 171.4 + 10v - 69.46 + 1908 dv/dt \text{ or}$$

$$F_e(t) = F(t) - F_{Ro} + F_o = F(t) - 171.4 + 69.46 = 10v + 1908 dv/dt \dots [6]$$

Equation [6] may be represented by the following block-diagram:



Question 52(d)

Taking the Laplace transform of the left and right-hand sides of equation [6] gives,

$$F_e(s) = 10 V(s) + 1908 sV(s) \dots [7]$$

Thus the transfer function, $G_v(s)$, relating car speed, $V(s)$ to the excess motive force, $F_e(s)$, when the car travels on a level road at speeds around $v_o = 50 \text{ km/h} = 13.89 \text{ m/s}$ under windless conditions is:

$$G_v(s) = \frac{V(s)}{F_e(s)} = \frac{1}{10 + 1908 s} \dots [8]$$